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Professor Sommer

Final Reflection

IDS 493

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As my time in the undergraduate Cybersecurity program at Old Dominion University gets closer to the end, I am able to look back and recognize my progress. Alongside getting ready to graduate in Spring 2026 with a 3.41 GPA, within this experience I have been challenged academically, professionally, and personally. I have grown in each of these categories to where I can reflect on my experiences alongside see how all of my interdisciplinary courses have shaped my career readiness. First, I believe there were many positive things I have gained from my coursework, such as improving my cybersecurity knowledge, as before my core classes I would consider myself to have a decent general understanding of cybersecurity, but not enough as someone who is getting their undergrad degree in the spring and planning to pursue a master's degree. Within this e-portfolio, I will show the experiences I've gained. Specifically, the three major skillsets: Vulnerability Assessment & Penetration Testing, Systems Administration & Scripting, and Cybersecurity Research & Consulting. By looking at nine different artifacts from my classroom assignments, real-world work, and service-learning internships, I can demonstrate how merging technical knowledge with human-centric communication is essential.

First, I believe there were many positive things I have gained from learning Vulnerability Assessment & Penetration Testing. The first artifact demonstrating my proficiency in this skillset is my Lab 5 assignment from CYSE 450: Ethical Hacking and Penetration Testing. The objective of the project was to provide hands-on experience on web security as well as SQL injection attacks. We also wanted to show how such attacks are executed by malicious parties in real settings. I downloaded and installed VirtualBox to access a safe environment to try and experiment with such attacks. The most challenging aspect was trying to plant an XSS trap by adding a new product that has a script in its name. I also solved database hurdles using SQL injection techniques like entering ’or’7’=’7’;# to bypass login screens.

Building on this, my second artifact is my experience generating a Tenable Vulnerability Scan Report. In my coursework, I learned how to use Tenable at a proficient level, analyzing and running scans on servers alongside generating reports of the data when I was working for the City of Suffolk via the Cyber Clinic. They taught us that regardless of the results, finding a vulnerability is only half the battle; knowing how to prioritize it is what actually matters. Research by Cheimonidis and Rantos (2025) emphasizes this, stating that “the future of vulnerability prioritization lies in the development of adaptive, context-aware metrics that leverage dynamic threat intelligence to enhance risk assessments” (p. 2). By understanding Tenable and looking at it from this perspective, it showed me that even if you’re not directly combating vulnerabilities, even just monitoring them is a huge step.

My final artifact for this first skill is my broader portfolio of Ethical Hacking Labs utilizing Kali Linux. This course introduces the basic terminologies used in ethical hacking and useful tools in relation to penetration testing on Kali Linux. I learned to explore the vulnerabilities in various systems and operate the industry-leading tools and framework to perform penetration testing on different target systems. Job advertisements heavily require this hands-on knowledge, and working directly within Kali Linux gave me the confidence to handle these tools in a real-world setting.

Next, I do believe another major area of growth for me was Systems Administration & Scripting. My first artifact for this second skill is my Python Cryptographic Calculator Project. For this project, I had to create a basic program using Python that could handle Basic Mod Calculations, Fermat's Little Theorem, and the Extended Euclidean Algorithm. Developing this script was a complex process because I had to create a menu function that shows options and welcomes the user, alongside a while loop to break. I had to use math functions to determine what will be the dividend and divisor and solve for quotient and remainder, alongside storing it in a tuple to iterate when back substituting until the remainder is 0. This artifact shows I am proficient in Python and can automate complex mathematical tasks.

Similarly, my Linux and Crontab Configuration assignments showcase my ability to use Linux at a knowledgeable level. To really secure a network, you have to understand how the operating system schedules and runs its tasks. I learned how to run scripts such as Metasploit and schedule recurring tasks using Crontab. By learning the command line and running tasks using subprocesses, I was able to bridge the gap between offensive security and everyday system maintenance, which is a connection I had not previously appreciated.

My third artifact for this skill is my real-world work experience at the ODU IT Help Desk. Since July 2025, I have been working resolving technical issues for EVMC and ODU faculty, staff, and students via ServiceNow. Tasks include using basic troubleshooting resources like knowledge bases or previous tickets alongside technical knowledge such as OS, network infrastructure, and Active Directory to assist and escalate tickets. Looking back from then to now, I am proud of myself for the steps I've taken, as getting an IT job at the school alongside my coursework would not have been possible unless I decided that it was time for me to finally apply myself toward my career. As Johnson (2025) explains, "automation removes the burden of repetitive tasks from IT staff so they can focus on critical activities like strategic planning and problem solving" (p. 5). By handling Active Directory and ServiceNow tickets, I am acting as the crucial human element that manages these systems.

The final skillset I have developed is Cybersecurity Research & Consulting. My seventh artifact overall is my CYSE 280 Research Paper titled "Usage of AI When Defending Cloud Based Services," where I received a 96/100 from Professor Gladden. Within this paper, I make an argument for why AI systems have proven to be successful when integrated to protect cloud-based services. Artificial intelligence refers to the ability of machines to mimic intelligent human behavior, such as learning, reasoning, problem-solving, and decision-making. In the context of OS management, AI algorithms can analyze data, identify patterns, and make automated decisions to optimize system performance. Writing this paper was a massive learning experience, as Kummara (2025) notes that the fusion of artificial intelligence with cloud safety produces strong, resilient, and scalable protection mechanisms that counter modern-day cybersecurity threats (p. 28).

Through researching this topic, I realized that my dream job is to work to help ensure AI is used ethically, as I believe AI is, if not already, the biggest emerging threat to secure systems.

My consulting work with the COVA CCI Cyber Clinic provided my next artifact: my Executive Reflection Presentation for the City of Suffolk. Through the clinic, under the professional guidance of Valor Cybersecurity and Mr. Greg Tomchick, our group worked with Dr. Kiriakou, Mr. Joshua Cox, and Mr. Miller alongside meeting with the networking team. Upon meeting with the clients for the first time, we established a schedule of meetings with them, with one visit on premise and the rest being weekly through Microsoft Teams. Alongside that, we established a form of communication where they preferred our emails to be encrypted to ensure confidentiality. Upon receiving feedback from the rough draft and dry runs, our group was able to improve our presentation to where we could successfully communicate remediation and mitigation strategies.

Finally, my Individual Final Reflection from my internship serves as my last artifact. Within my time in this program, I've noticed that I can be a perfectionist, especially when it comes to things I am passionate about such as cybersecurity. However, working alongside Carla, who really helped us when it came to working with Suffolk and their expectations, made the presentation feel far less overwhelming. Consulting requires looking at the human element of security. Research by Charitoudi and Blyth (2013) supports this, stating that "a more social oriented model must be developed that views an organization as a holistic construct comprising of people and technology" (p. 33). This clinic has provided me with an amazing opportunity that has set me up for success within my career.

Reflecting on my coursework, internships, and artifacts as a whole, it is clear that interdisciplinary methods and theories were foundational to my understanding of cybersecurity. Foundational courses such as IDS 300W were pivotal in preparing me. It taught me that cybersecurity is not just about computers; it involves business communication and ethical reasoning. Whether I was working in VirtualBox running SQL injections, writing Python scripts for cryptography, or encrypting emails for the City of Suffolk, I had to approach every problem from multiple angles. As Jacob et al. (2018) state, “an equal amount of knowledge outside the technical domain is pivotal to understand sophisticated challenges in cybersecurity.” Ultimately, being an interdisciplinary thinker is essential in cybersecurity because threats span networks, policies, and human behavior. The impact of these experiences will reflect on the rest of my time here at ODU as I transition into my master’s degree. All together, this portfolio has provided me with proof that I am ready to tackle the multifaceted challenges of the cybersecurity industry.

References

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